

Assignment 2: Binary Arithmetic and Multiplication

1) Perform the following addition operations. All values are 8-bit 2's Complement representation.

a) $(45)_{10} + (47)_{10}$
 $0010\ 1101 + 0010\ 1111$

			1			1	1	1
0	0	1	0		1	1	0	1
0	0	1	0		1	1	1	1
0	1	0	1		1	1	0	0

or $(92)_{10}$

b) $(21)_{10} + (127)_{10}$
 $0001\ 1111 + 0111\ 1111$

			1	1	1	1		
0	0	0	1		1	1	1	1
0	1	1	1		1	1	1	1
1	0	0	1		1	1	1	0

or $(158)_{10}$

c) $(53)_{10} + (10)_{10}$
 $0011\ 0101 + 0000\ 1010$

			0	0	1	1		0	1	0	1
			0	0	0	0		1	0	1	0
			0	0	1	1		1	1	1	1

or $(63)_{10}$

2) Perform the following Conversions

a) Decimal Value -95 to binary using 1's Complement:

95 / 2 = 47 R1
47 / 2 = 23 R1
23 / 2 = 11 R1
11 / 2 = 5 R1
5 / 2 = 2 R1
2 / 2 = 1 R0
1 / 2 = 0 R1

0

$0101\ 1111\ (-95)_2$

$1010\ 0000$ flip bits

(1's Complement) $\nearrow$

b) Decimal Value -95 to binary using 2's Complement:

$1010\ 0000$

+ 1 Add 1

$1010\ 0001$

(2's Complement) $\nearrow$

c) 2's Complement binary Value 1011 1000 to decimal.

if 1, negative
if 0, positive

- $1011\ 1000$

flip bits (1's)

Add 1 (2's)

$0100\ 1000$

convert to decimal Value

128	64	32	16	8	4	2	1
0	1	0	0	1	0	0	0

$64 + 8 = (72)_{10}$

3) Perform the following Subtractions. All values are 8-bit 2's Complement Representation, the result should also be in 8-bits

a) $(0) - (-93)$
 $0000\ 0000 - 0101\ 1101 = 1010\ 0011$ or $(-93)_{10}$

b) $(247) - (26)$
 $1111\ 0111 - 0101\ 0110$

Add 1

$1111\ 0111 + 1010\ 1001$

$1111\ 0111 + 1010\ 1010$

$1010\ 0001$ or $(161)_{10}$

c) $(112) - (208)$
 $0111\ 0000 - 1101\ 0000$

Add 1

$0111\ 0000 + 0010\ 1111$

$0111\ 0000 + 0011\ 0000$

$1010\ 0000$ or $(-96)_{10}$

4) Multiply $0111\ 0111$ and 11 in binary. Do not worry about overflow for this problem.

0	1	1	1		0	1	1	1
x								1
	0	1	1	1		0	1	1
	0	1	1	1		0	1	1
	0	1	1	1		0	1	1
	1	0	1	1		0	1	0

(119) $\times$ (3)

$10110\ 0101$

$(357)_{10}$